

ENTHUSIAST, LEADER & ACHIEVER COURSE

ALL PHASE

TARGET : PRE-MEDICAL 2020

Test Type : **MAJOR**

Test Pattern : **NEET(UG)**

TEST DATE : 31 - 03 - 2020

TEST SYLLABUS : 04, 05 & 06

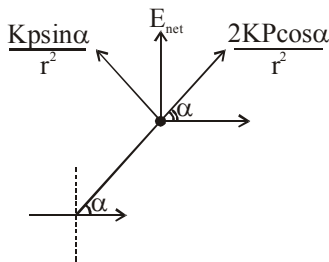
Q	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A	4	3	3	2	4	4	4	2	4	4	2	4	1	2	2	4	3	3	2	1	4	3	2	2	2	2	1	4	4	2
Q	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A	4	3	4	2	1	3	1	2	1	3	4	3	4	4	1	1	2	1	3	1	4	2	1	2	4	1	3	2	3	1
Q	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A	4	2	1	1	1	4	1	4	2	1	1	4	3	2	3	4	3	1	2	3	4	3	3	3	1	3	2	2	3	1
Q	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A	2	1	1	1	1	1	1	2	2	4	2	4	4	2	4	3	1	3	3	3	4	1	1	2	3	3	3	1	3	4
Q	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A	3	1	1	2	3	4	2	3	1	1	2	2	3	2	4	4	2	4	4	4	2	2	4	4	2	3	2	4	3	2
Q	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A	1	1	3	2	3	2	3	3	3	2	3	3	3	1	4	4	3	3	3	1	2	4	3	2	3	1	2	3	2	2

HINT - SHEET

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|---|---|
| <ol style="list-style-type: none"> NCERT XII Pg # 149 NCERT XII Pg # 153 Module 1 Pg # 90 NCERT XII Pg # 136
 $P = 0.6, q = 1 - P = 0.4$
 $Aa \Rightarrow 2pq \Rightarrow 2 \times 0.6 \times 0.4 = 0.48$
 so, no. of individuals = $48 \times 100 = 48$ NCERT XII Pg # 231, para 13.2.2 NCERT XII Pg # 261, 262 (E), 286 (H) NCERT-XII, Pg#221, para-13.1 NCERT XII, Page # 35, Para-2.4.1 NCERT XII Pg # 64, para 1, Pg#57(E), Para-1 NCERT XII Pg # 169 NCERT XII Pg # 229, para 13.2.2 NCERT XII Pg # 263 (E), 287 (H) NCERT-XII, Pg#262 NCERT XII, Page # 35, Para-2.4.2, Fig. 2.14 NCERT XII Pg # 155 NCERT XII Pg # 150 NCERT XII Pg # 127 | <ol style="list-style-type: none"> NCERT XII Pg # 168 NCERT XII Pg # 227, para 13.2.1 Module 3 Pg # 36 NCERT-XII, Pg#229, para-13.2.2 NCERT XII Pg # 46 (E), para 3, Pg#49(H) NCERT XII Pg # 131 NCERT XII Pg # 169 Module 3 Pg # 3 NCERT XII Pg # 271 (E), 295 (H) Cyprus $\rightarrow$ 1 MMC $\xrightarrow{\text{meiosis}}$ 1 microspore (M) NCERT XII Pg # 63 NCERT XII Pg # 49, para 4, Pg#46(E), Para-3 NCERT XII Pg # 132 NCERT XII Pg # 235-236, para 13.2.4 Module 3 Pg # 3 NCERT XII Pg # 273 (E), 297 (H)
 Emission standards are being updated from time to time
 समय-समय पर उत्सर्जन मानकों का नवीनीकरण किया जाता है। |
|---|---|

47. NCERT-XII, Pg#24, para-2.2.2
51. NCERT XII Pg # 160
52. NCERT XII Pg # 49, fig. 3.8, Pg#53(H)
53. NCERT XII Pg # 135
55. NCERT XII Pg # 232 (E), 252 (H)
56. NCERT-XII, Pg#279 (E), Pg#306 (H)
57. NCERT-XII, Pg#36, para-2.4.3
61. NCERT XII Pg # 161
62. NCERT XII Pg # 55, para 1, Pg#50(E)
64. NCERT XII Pg # 220, para 13.1
65. NCERT XII Pg # 222 (E), 242 (H)
66. NCERT-XII, Pg#279 (E), Pg#304 (H)
67. NCERT-XII, Pg#27, para-2.2.2
72. NCERT XII Pg # 55, para 1, Pg#51(E)
73. NCERT XII Pg # 131
74. NCERT XII Pg # 264, para 15.1.4
75. NCERT XII Pg # 220 (E), 239 (H), fig 13.1
76. NCERT-XII, Pg#250, 251
77. NCERT XII, Page # 28, para-2.2.3
82. NCERT XII Pg # 59, para 1, Pg#54(E)
83. NCERT XII Pg # 129
84. NCERT XII Pg # 260, fig. 15.1
85. NCERT XII Pg # 259 (E), 283 (H)
86. NCERT, Pg#223, para-13.1.2
87. NCERT XII, Page # 34, Para-2.3
91. E_{net} along y-axis

$$\Rightarrow \frac{Kp \sin^2 \alpha}{r^2} = \frac{2Kp \cos^2 \alpha}{r^2}$$



$$\Rightarrow \tan \alpha = \sqrt{2} \Rightarrow y/x = \sqrt{2} \Rightarrow y = 2m$$

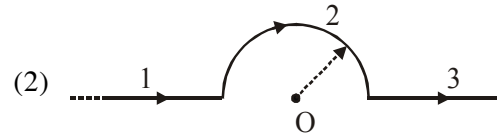
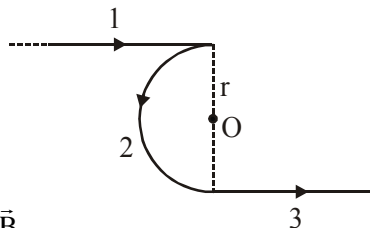
92.(1) $B_1 = \frac{\mu_0 I}{4\pi r} = B_3$

$$\odot B_2 = \frac{\mu_0 I}{4r}$$

$$\vec{B}_{\text{net}} = \vec{B}_1 + \vec{B}_2 + \vec{B}_3$$

$$B_{\text{net}} = \frac{\mu_0 I}{4r} \odot$$

.....(i)



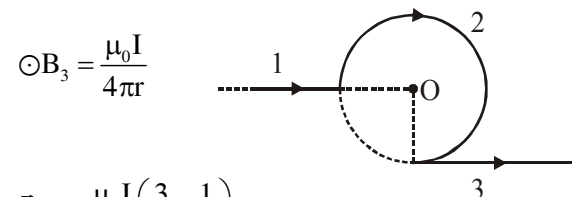
$$B_1 = B_3 = 0$$

$$\vec{B}_{\text{net}} = \vec{B}_1 + \vec{B}_2 + \vec{B}_3$$

$$\otimes B_2 = \frac{\mu_0 I}{4R} = \frac{\mu_0 I}{4r} \otimes \dots\dots(ii)$$

(3) $B_1 = 0$

$$\otimes B_2 = \frac{\mu_0 I}{4\pi r} \left(\frac{3\pi}{2} \right) = \left(\frac{3\mu_0 I}{8r} \right)$$



$$\vec{B}_{\text{net}} = \frac{\mu_0 I}{4r} \left(\frac{3}{2} - \frac{1}{\pi} \right) \otimes \dots\dots(iii)$$

$$B_{(1)} : B_{(2)} : B_{(3)}$$

$$-1 : 1 : \left(\frac{3}{2} - \frac{1}{\pi} \right)$$

$$\Rightarrow \frac{-\pi}{2} : \frac{\pi}{2} : \left(\frac{3\pi}{4} - \frac{1}{2} \right)$$

94. $I = 4I_0 \cos 2\left(\frac{\phi}{2}\right)$

$$I_0 = 4I_0 \cos^2\left(\frac{\phi}{2}\right)$$

$$\text{or } \cos\left(\frac{\phi}{2}\right) = \frac{1}{2}$$

$$\text{or } \frac{\phi}{2} = \frac{\pi}{3}$$

$$\text{or } \frac{\phi}{2} = \frac{\pi}{3}$$

$$\text{or } \phi = \frac{2\pi}{3} = \frac{2\pi}{\lambda} \Delta x$$

$$\text{or } \frac{1}{3} = \frac{1}{\lambda} \left(\frac{yD}{D} \right)$$

$$\therefore y = \frac{\lambda}{3\left(\frac{d}{D}\right)} = \frac{6 \times 10^{-7}}{3 \times 10^{-4}} = 2 \times 10^{-3} \text{ m} = 2 \text{ mm}$$

95. $E = \Delta m \times C^2$
 $= [m_n - (m_p + m_e)] \times C^2$
 $= [1.6747 \times 10^{-27} - (1.6725 \times 10^{-27} + 0.00091 \times 10^{-27})] \times C^2$
 $= 0.73 \text{ MeV}$

96. $V_{10} - V_{20} = -\int_{20}^{10} \vec{E} \cdot d\vec{x}$

$$V_{10} - V_{20} = -\int_{20}^{10} \left(\frac{100}{x^2}\right) dx = 100 \left[\frac{1}{x}\right]_{20}^{10}$$

$$= 100 \left[\frac{1}{10} - \frac{1}{20}\right] = 5 \text{ volt}$$

97. $B = \mu_0 n i$
 $i = \frac{B}{\mu_0 n} = \frac{20 \times 10^{-3} \times 10^{-2}}{4\pi \times 10^{-7} \times 20} \approx 8 \text{ A}$

98. $R = \frac{V}{i} = \frac{120}{0.5} = 240 \Omega$

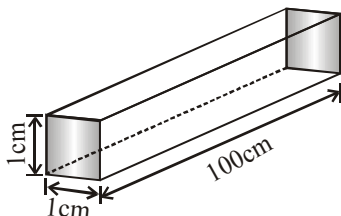
100. $T_a = T$

$$\left(\frac{1}{2}\right)^2 = \frac{25}{100} N_0 = N_0 \left(\frac{1}{2}\right)^{\frac{t}{T_n}}$$

$$2 = \frac{t}{T_n} \quad \left\{ T_a = \frac{1}{\lambda} \text{ \& } T_n = \frac{\ln 2}{\lambda} \right\}$$

$$t = 2T_n = \frac{2 \ln 2}{\lambda} = 2T \ln 2$$

101. Length $l = 1 \text{ cm} = 10^{-2} \text{ m}$

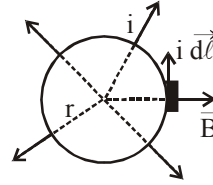


Area of cross-section $A = 1 \text{ cm} \times 100 \text{ cm}$
 $= 100 \text{ cm}^2 = 10^{-2} \text{ m}^2$

Resistance $R = 3 \times 10^{-7} \times \frac{10^{-2}}{10^{-2}} = 3 \times 10^{-7} \Omega$

102. Force on small element $dF = I d\ell B \sin \theta$
 So, force on complete loop

$$F = \int i d\ell B \sin \theta$$



Here $\theta = 90^\circ = \text{constant} \Rightarrow F = iB \int \ell$
 $F = iB 2\pi r$

103. $i = a + b \sin \omega t$
 $i^2 = a^2 + b^2 \sin^2 \omega t + 2ab \sin \omega t$
 $(i^2) = (a^2) + (b^2 \sin^2 \omega t) + (2ab \sin \omega t)$

$$(i^2) = a^2 + \frac{b^2}{2}$$

$$\sqrt{(i^2)} = \sqrt{a^2 + \frac{b^2}{2}} = I_{\text{rms}}$$

104. $\frac{hc}{\lambda} = W_0 + K_{\text{max}} \Rightarrow \frac{hc}{\lambda_A} = W_0 + K_A \quad \dots(i)$

and $\frac{hc}{\lambda_B} = W_0 + K_B \quad \dots(ii)$

Subtracting (i) from (ii),

$$hc \left[\frac{1}{\lambda_B} - \frac{1}{\lambda_A} \right] = K_B - K_A$$

$$\Rightarrow hc \left[\frac{1}{\lambda_B} - \frac{1}{2\lambda_B} \right] = K_B - K_A \Rightarrow \frac{hc}{2\lambda_B} = K_B - K_A \quad \dots(iii)$$

From (ii) and (iii), $2K_B - 2K_A = W_0 + K_B$

$$\Rightarrow K_B - 2K_A = W_0$$

$$\Rightarrow K_A = \frac{K_B - W_0}{2} \text{ which gives } K_A < \frac{K_B}{2}$$

106. $V = \varepsilon - ir = \varepsilon - \frac{\varepsilon r}{r + nr} = \varepsilon - \frac{\varepsilon}{n+1}$

$$V = \frac{n\varepsilon}{n+1} \Rightarrow \frac{V}{\varepsilon} = \frac{n}{n+1}$$

107. Magnetic field at centre is zero

$$B_1 = B_2$$

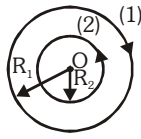
$$\frac{\mu_0 N_1 I_1}{2R_1} = \frac{\mu_0 N_2 I_2}{2R_2}$$

Given that

$$(I_1 = I_2, N_1 : N_2 = n_1 : n_2, R_1 : R_2 = 2 : 1)$$

$$\frac{N_1}{N_2} = \frac{I_2}{R_2} \times \frac{R_1}{I_1} \Rightarrow \frac{n_1}{n_2} = \frac{R_1}{R_2} (\because I_1 = I_2)$$

$$\frac{n_1}{n_2} = \frac{2}{1}$$



108. $f_0 = \frac{1}{2\pi\sqrt{LC}}$

$$\therefore \epsilon_r = 16$$

$$\therefore C' = 16C \Rightarrow f' = \frac{1}{2\pi\sqrt{L16C}} = \frac{f_0}{4}$$

109. By using $E = W_0 + K_{\max}$

$$E = \frac{12375}{5000} = 2.475 \text{ eV and } K_{\max} = eV_0 = 1.36 \text{ eV}$$

$$\text{So } 2.475 = W_0 + 1.36 \Rightarrow W_0 = 1.1 \text{ eV.}$$

110. $A_V = \beta \left(\frac{R_L}{R_1} \right) = 50 \times \left(\frac{5000}{2000} \right) = 125$

$$\therefore \Delta V_e = A_V \times \Delta V_b = 125 \times 10 \text{ mV} = 1.25 \text{ volt.}$$

111. $E = \frac{e}{(R + R_h + r)} \cdot \frac{R}{L} \times I$

$$= \frac{2}{(10 + 40 + 0)} \times \frac{10}{1} \times 0.4 = 0.16 \text{ V}$$

112. $\therefore \tan \theta = \frac{B_V}{B_H}$

$$\Rightarrow B_V = B_H \tan \theta$$

$$= 0.6 = \frac{4}{3} = 0.8 \text{ G}$$

113. $X_L = \omega L = 0.4 \times 2\pi \times \frac{30}{\pi} = 24\Omega$

$$Z = \sqrt{10^2 + 24^2} = 26\Omega$$

$$P_{\text{avg}} = V_{\text{rms}} I_{\text{rms}} \cos \phi = V_{\text{rms}} \left(\frac{V_{\text{rms}}}{Z} \right) \left(\frac{R}{Z} \right)$$

$$= \frac{RV_{\text{rms}}^2}{Z^2} = \frac{5}{8} \text{ W}$$

114. Momentum of electron $p_e = \sqrt{2meV}$

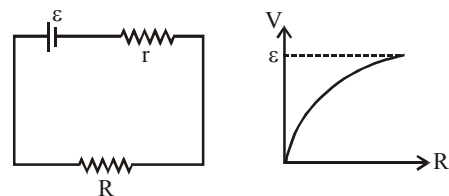
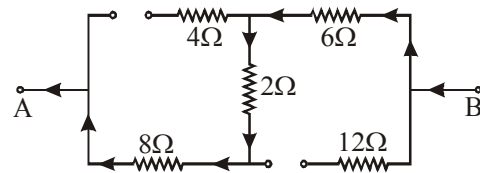
Momentum of proton $p_p = \sqrt{2MeV}$

$$\therefore \frac{\lambda_p}{\lambda_e} = \frac{h/p_p}{h/p_e} = \frac{p_e}{p_p} = \frac{\sqrt{2meV}}{\sqrt{2MeV}} = \sqrt{\frac{m}{M}}$$

$$\therefore \lambda_p = \lambda_e \sqrt{\frac{m}{M}} = \lambda \sqrt{\frac{m}{M}}$$

115. According to the given figure A is at lower potential w.r.t. B. Hence both diodes are in reverse biasing, so equivalent circuit can be redrawn as follows.
 $\Rightarrow$ Equivalent resistance between A and B

$$R = 8 + 2 + 6 = 16 \Omega$$



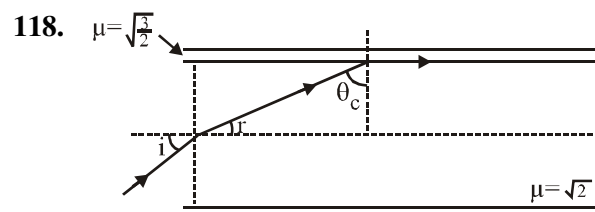
116.

$$V = \epsilon - ir$$

$$V = \frac{\epsilon R}{R + r}$$

$$V = \frac{\epsilon}{1 + \frac{r}{R}}$$

117. $r = \frac{\sqrt{2mk}}{qB} = \frac{1}{B} \sqrt{\frac{2mV}{q}} \Rightarrow r \propto \sqrt{m} \Rightarrow \frac{m_1}{m_2} = \left(\frac{R_1}{R_2} \right)^2$



Ist method

$$\sin \theta_c = \frac{\mu_R}{\mu_d} = \frac{\sqrt{2}}{\sqrt{3}} \quad \sin \theta_c = \frac{\sqrt{3}}{2}$$

$$\theta_c = 60^\circ$$

$$r = 90^\circ - \theta_c = 90^\circ - 60^\circ$$

$$r = 30^\circ$$

$$\text{Now, } 1 \times \sin i = \sqrt{2} \times \sin(30^\circ)$$

$$i = 45^\circ$$

IInd method

$$\sin i = \sqrt{n_1^2 - n_2^2} = \sqrt{(\sqrt{2})^2 - \left(\frac{\sqrt{3}}{2}\right)^2} = \frac{1}{\sqrt{2}}$$

$$\Rightarrow i = 45^\circ$$

$$119. \quad \lambda_1 = 1 \text{ \AA} \Rightarrow v_1 = \frac{150}{(1)^2} = 150v$$

$$kE_1 = 150 \text{ eV}$$

$$\lambda_2 = \frac{1}{2} \text{ \AA} \Rightarrow v_2 = \frac{150}{\left(\frac{1}{2}\right)^2} = 600v$$

$$kE_2 = 600 \text{ eV}$$

$$\Delta K = KE_2 - KE_1$$

$$= 600 - 150 = 450 \text{ eV}$$

120. Current I in $4k\Omega$ resistance

$$I = \frac{60-10}{4000} = 12.5 \text{ mA}$$

Current in $2k\Omega$ resistance is

$$I_L = \frac{10}{2000} = 5 \text{ mA}$$

and

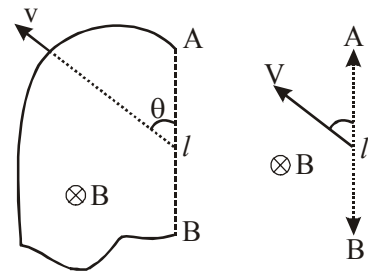
$$\begin{aligned} I_Z &= I - I_L \\ &= (12.5 - 5) \text{ mA} \\ &= 7.5 \text{ mA} \end{aligned}$$

$$121. \quad \text{Energy density} = \frac{1}{2} \epsilon_0 E^2$$

$$= \frac{1}{2} \times 10^{-11} \times \left(\frac{10}{5 \times 10^{-2}} \right)^2$$

$$= 2 \times 10^{-7} \text{ J/m}^3$$

122. The same emf will be induced in the straight imaginary wire joining A and B, which is $Bvl \sin \theta$.



123. 40 cm from pole in the medium of refractive index

124. According to Moseley's Law :

$$\sqrt{v} = a(Z-1)$$

$$\Rightarrow (Z-1)^2 \propto v$$

$$\text{or } (Z-1)^2 \propto 1/\lambda$$

$$\Rightarrow$$

$$\frac{(Z_{Mo}-1)^2}{(Z_{Cu}-1)^2} = \frac{\lambda_{Cu}}{\lambda_{Mo}} \text{ or } \lambda_{Cu} = \lambda_{Mo} \frac{(Z_{Mo}-1)^2}{(Z_{Cu}-1)^2}$$

$$= 0.71 \times \left(\frac{41}{28} \right)^2 = 1.52 \text{ \AA}$$

$$125. \quad \Delta I_C = \beta \times \Delta I_B = 50 \times 40 \times 10^{-6} = 2 \text{ mA}$$

$$\begin{aligned} \Delta V_0 &= \Delta I_C R_C = (2 \times 10^{-3}) \times 3 \times 10^3 \\ &= 6 \text{ V} \end{aligned}$$

$$126. \quad C_0 = \frac{\epsilon_0 A}{d} = 18 \quad \dots(1)$$

$$C = \frac{K \epsilon_0 A}{3d} = 72 \quad \dots(2)$$

From (1) & (2)

$$\frac{K}{3} = \frac{72}{18}$$

$$K = 12$$

127. By Lenz's law, induced effects always oppose the cause of induction. So when the first loop is moves towards the smaller loop it will be repelled by the induced current in the loop.

$$128. \quad \frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

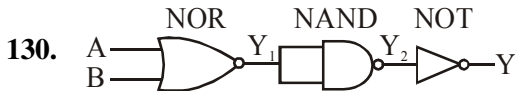
$$u = -\frac{x}{2}, v = -x$$

$$\frac{1}{f} = -\frac{1}{x} + \frac{2}{x}$$

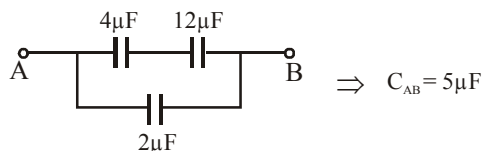
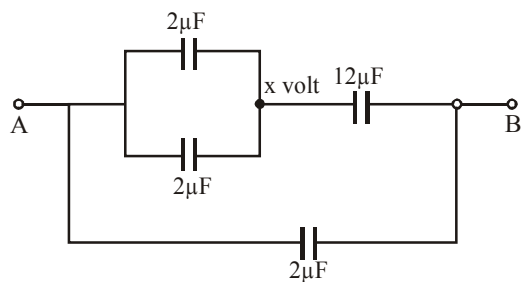
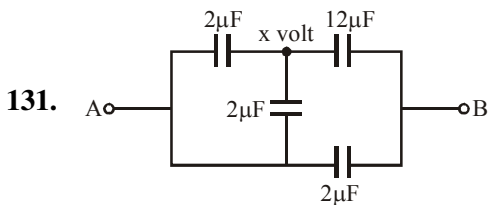
$$\frac{1}{f} = \frac{1}{x}$$

$$f = 1\text{m}$$

129. β -is emitted by conversion of neutron into proton.



NOR			NAND			NOT		
A	B	Y ₁	Y ₁	Y ₁	Y ₂		Y	
			A	B				
0	0	1	1	1	0		1	
0	1	0	0	0	1		0	
1	0	0	0	0	1		0	
1	1	0	0	0	1		0	



132.
$$E = \frac{R/2}{2} \frac{dB}{dt}$$

$$= \frac{2.5 \times 10^{-2}}{4} \cdot (6 \times 4 + 24)$$

$$F = \frac{1.6 \times 10^{-19} \times 2.5 \times 10^{-2} \times 48}{4}$$

$$= 48 \times 10^{-21} \text{ N}$$

133. Here, $u_0 = -200 \text{ cm}$, $f = 30 \text{ cm}$

$$\frac{1}{v_0} + \frac{1}{200} = \frac{1}{30}$$

or
$$\frac{1}{v_0} = \frac{1}{30} - \frac{1}{200}$$

$$\therefore v_e = \frac{600}{17} = 35.3 \text{ cm}$$

Now, $L = v_0 + f_e = 35.3 + 3 = 38.3 \text{ cm}$.

134. $P = \frac{NE}{t} \Rightarrow 300 \times 10^6 = \frac{N \times 170 \times 10^6 \times 1.6 \times 10^{-19}}{t}$

$$\therefore \text{Number of atoms per sec } \frac{N}{t} = 1.102 \times 10^{19}$$

$$\text{Number of atoms per hour} = 1.02 \times 10^{19} \times 3600 = 3.97 \times 10^{22}$$

136. $t_{1/2} \propto [A]^{1-n} \therefore n = 1$

137. moles of hydrogen gas = $\frac{224}{22400} = 10^{-2}$

$$g_{eq} \text{ to } H_{2(g)} = \text{mole} \times n \text{ factor} = 2 \times 10^{-2}$$

$$g_{eq} = \frac{i \times t}{96500}$$

$$2 \times 10^{-2} = \frac{i \times 3600}{96500}$$

$$i = 0.536 \approx 0.54 \text{ A}$$

138. Fact

139. Fact

140. Thermite mixture

$\Rightarrow$ metal oxide + Al

(3 : 1)

141. $t_{1/2} = \frac{0.693}{K}$

$$t_{1/2} = \frac{0.693}{0.69 \times 10^{-1}} = 10 \text{ min}$$

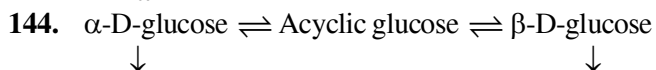
$$t_{1/2} = 10 \times 60 \text{ sec} \Rightarrow 600 \text{ sec}$$

142. $E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{0.06}{n} \log Q$

$\therefore E_{\text{cell}} = (1.66 - 0.40) - \frac{0.06}{6} \log \frac{[\text{Al}^{+3}]^2}{[\text{Cd}^{+2}]^3}$

$E_{\text{cell}} = 1.26 - 0.01 \log \frac{(0.1)^2}{(0.01)^3}$

$E_{\text{cell}} = 1.22 \text{ volt}$



specific rotation $112.2^{\circ} \rightarrow \boxed{52.5^{\circ}} \leftarrow$ specific rotation 18.7°

equilibrium value
(Mutarotation)

145. at higher temp. reaction would be feasible

147. Case-1

$P^{\circ} - P_s = 10 \Rightarrow X_B = 0.2$

$\text{RWP} = X_B$

$\Rightarrow \frac{10}{P^{\circ}} = 0.2 \Rightarrow 50$

Case-2

$P^{\circ} - P'_s = 20 \Rightarrow X_A = ?$

Now, $\frac{20}{P^{\circ}} = X_B$

$\Rightarrow \frac{20}{50} = X_B$

$\Rightarrow 0.4 = X_B$

$\Rightarrow X_A = 0.6$

148. Reactivity of C-H bond = $3^{\circ} > 2^{\circ} > 1^{\circ}$

149. Fact

151. Gold no. $\propto \frac{1}{\text{Protective power}}$

152. $P_T = P_A^{\circ} X_A + P_B^{\circ} X_B$

$500 = P_A^{\circ} \times \frac{1}{3} + P_B^{\circ} \times \frac{2}{3}$

$\Rightarrow 1500 = P_A^{\circ} + 2P_B^{\circ} \dots (i)$

$525 = P_A^{\circ} \times \frac{1}{4} + P_B^{\circ} \times \frac{3}{4}$

$\Rightarrow 2100 = P_A^{\circ} + 3P_B^{\circ}$

$\Rightarrow P_B^{\circ} = 600 \text{ \& } P_A^{\circ} = 300$

153. Fact

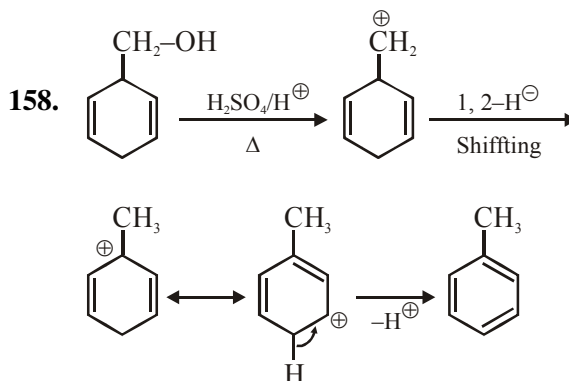
157. Property of a binary ideal solution

$\Delta H_{\text{mix}} = 0$

$\Delta V_{\text{mix}} = 0$

$\Delta G_{\text{mix}} < 0$

$\Delta S_{\text{mix}} > 0$



$0.4 \quad 1 \quad 0$

$0.1 \quad 0.4 \quad 0.3$

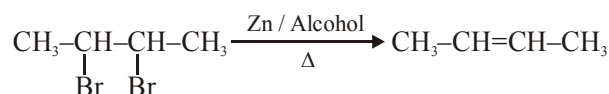
$r = k p_A^1 p_B^2$

$r_0 = k(0.4)^1(1)^2 \dots (i)$

$r_t = k(0.1)^1(0.4)^2 \dots (ii)$

$\frac{r_t}{r_0} = \frac{0.4 \times 0.4 \times 0.1}{0.4} = 0.04 = \frac{4}{100} = \frac{1}{25}$

163.



167. Fact

168. Fact

172. 1° -amine give foul smell when heated with CHCl_3 and KOH .

